

NutrioBox Dispatch — Auto-Post Logic & Calculations

How the gun's auto-post, delta calculations, ledgers and Submit work — and why the ledger must match PACT.

1 The core idea: auto-post sends DELTAS, not the whole order

Auto-post runs on a timer (every **10 minutes**) while an order is open. On each cycle it does not re-send the whole order — it sends only the items **newly scanned since the last successful post**. That newly-scanned amount is called the **delta**. This lets the floor keep scanning while items dispatch in the background.

Each cycle performs four steps:

- **Autosave** the scanned list to the device.
- **Refresh stock** from PACT (so quantities are current).
- **Compute the delta** = scanned so far – already posted.
- **Post the delta** to PACT; on confirmed success, record those packs as 'posted' so they are never sent again.

2 The delta calculation

For every product+batch line, the amount sent on a cycle is:

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delta(line) = scanned(line) - alreadyPosted(line)
```

scanned = total packs scanned for that line so far. **alreadyPosted** = packs already confirmed onto the order's voucher (tracked in the ledgers, section 4). Only lines with delta > 0 are sent. Each line is also capped at that batch's available stock.

3 First, second and third auto-post — worked example

Say the operator keeps scanning a single product over 30 minutes. Here is what each cycle sends:

Cycle	Scanned so far	Already posted	Delta sent this cycle	Voucher action
1st auto-post (t=10 min)	8	0	8 (8 – 0)	CREATE voucher with 8
2nd auto-post (t=20 min)	14	8	6 (14 – 8)	APPEND 6 to same voucher
3rd auto-post (t=30 min)	19	14	5 (19 – 14)	APPEND 5 to same voucher
Submit (final)	22	19	3 (22 – 19)	Post remaining 3, mark COMPLETED

Key point: the three cycles build **one growing voucher** (8, then +6, then +5 = 19), and Submit adds the last 3 to reach 22. Total delivered = 22 = total scanned. Nothing is sent twice.

4 What 'already posted' means — the two ledgers

The whole delta system depends on an accurate memory of what has already been dispatched. Two ledgers hold it:

- **On the gun (postedPacks):** updated only when an auto-post is *confirmed* by PACT. Reset when the order is reopened.
- **On the worker (voucher-map.json):** the source of truth for what is already on each order's single voucher. Even if the gun re-sends everything, the worker posts only packs not already on the voucher — this is the guard against double-dispatch.

On top of both, **PACT itself caps every line at its own pending quantity** — it can never deliver more than what was ordered, whatever gets sent.

5 Submit (manual) vs auto-post

- **Auto-post** = background, every 10 min, sends the delta, keeps the order open, appends to the same voucher.
- **Submit** = the *final* post. It sends all remaining un-posted scanned items in one go, then marks the order **COMPLETED** and **stops the auto-post timer**. Use it when the dispatch is finished.
- If an auto-post ends without a confirmed success, auto-post **pauses** for safety (a stock move is never repeated blindly) until the operator verifies in PACT.

6 Why an order can get 'stuck' (the failure mode)

The delta logic is correct **only if** the 'already posted' memory matches what PACT actually delivered. It drifts when a post is *recorded* as done but PACT took fewer units (a partial or a commit that never finalised). Then:

- The ledger over-counts — it thinks lines are fully posted that PACT still shows as pending.
- Every later delta = scanned – (too-high) posted ≈ 0 , so those items are never re-sent.
- The gun shows them **green** ('scanned / done') while PACT still lists them **pending** — and the order can even show COMPLETED while PACT still owes stock.

Fix (principle): drive both the green/pending status *and* the delta from PACT's **live delivered quantity** (ordered – pending), not from the gun's own memory. Then 'posted' always equals what PACT actually has, a re-submit always sends exactly what is still pending, and no order can look done while PACT is still owed stock.

Reference sheet for the NutrioBox / Sukhna Foods dispatch team. Numbers in section 3 are an illustrative worked example.